

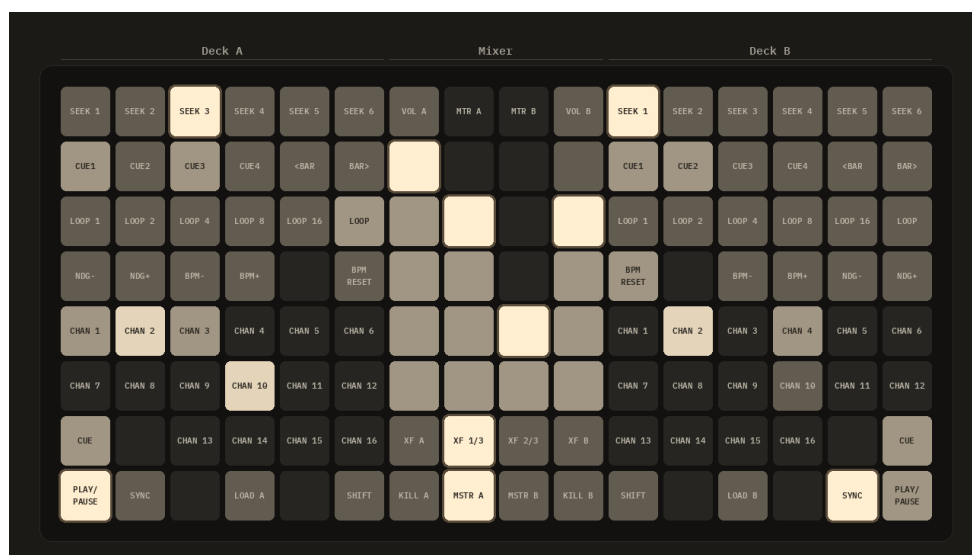
# MIDARTO

A two-deck MIDI file DJ for monome norms

User guide · version 2.14

Created by Glenn Jones, Cutie Suzuki & DJ FingaBlasta

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Load .mid files into two decks, mix them with a crossfader and play them to a MIDI sound module or to nb voices. Everything core works from the norms alone; a 16x8 grid adds cues, loops, channel mutes, sync and big transport keys.

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## 1 Install

You need a norns running a recent system update, and a MIDI sound module connected over USB MIDI. A General MIDI module is ideal, because most MIDI files are written for General MIDI sounds.

- **Unzip** midarto.zip. You get a folder called midarto containing midarto.lua, lib/, demo/ and a README.
- **Copy** the whole midarto folder into dust/code/ on your norns. Either open the norns network share from a computer on the same network (smb://norns.local on macOS, \\norns.local on Windows; user we, password sleep unless you changed it), or use a terminal:

```
scp -r midarto we@norns.local:~/dust/code/
```

- **Run it:** on the norns, go to SELECT and choose MIDARTO. A 5-second splash screen shows an 8-bit DJ at the decks next to the title, then the credits; press any key, knob or grid key to skip it.

The first run creates a songs folder, dust/data/midarto/songs/, and puts two demo songs in it: acid\_trk.mid (120 BPM) and breaks.mid (98.5 BPM).

### Your own files

Fill the songs folder, dust/data/midarto/songs/, with your own .mid or .midi files. Subfolders are fine, for example one per set or genre. The browser always opens in the songs folder, and you can go up to browse anywhere inside dust. You can delete the demo songs once you've added your own; if the folder is ever empty, Midarto puts them back.

**Upgrading from v2.4 or earlier:** songs you kept directly in dust/data/midarto/ are moved into the songs folder the first time v2.5 runs.

### Optional: nb voices

To play a deck through nb voices instead of MIDI, install the nb library (dust/code/nb) and any nb voice mods, then restart Midarto. An *nb voice* output and a voice selector then appear in each deck's parameters. Without nb, Midarto offers MIDI output only.

## 2 First setup

Open PARAMS > EDIT and scroll to MIDARTO. Each deck has its own section, followed by the mix settings. Midarto saves them a couple of seconds after you change them, so they survive a power cut, and keeps them when you update to a new version. If an update needs to adjust an old setting, it does so once and shows what it changed.

| Parameter            | What it does                                                                                                                                                                     |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| midi outputs         | <b>one device</b> (default): both decks play to deck A's MIDI device. <b>two devices</b> : each deck has its own.                                                                |
| output               | <b>midi</b> or <b>nb voice</b> (nb only if installed).                                                                                                                           |
| midi device          | Which MIDI port the deck plays to (1–16, with device names). With one device only deck A's shows; with two, deck B defaults to port 2.                                           |
| channel shift        | Moves the file's channels up by this amount, so two decks don't fight over channels. Deck A defaults to 0, deck B to 8.                                                          |
| drums stay on 10     | Keeps channel 10 drums on 10 when shifting; other channels skip 10.                                                                                                              |
| nb voice             | The voice used when output is nb voice.                                                                                                                                          |
| crossfader curve     | <b>dj</b> : both decks full at the centre. <b>smooth</b> : constant power.                                                                                                       |
| cue to first note    | <b>first note</b> (default), <b>first drum</b> or <b>off</b> . On load, hot cue 1 and the deck go to the start of the bar where the music begins, skipping silence at the start. |
| start on bar         | <b>on + sync</b> (default), <b>on</b> or <b>off</b> . PLAY on a stopped deck waits for the other deck's next bar line while it plays; on + sync also locks the tempo.            |
| quantize jumps       | Seek, cue and bar jumps wait for the next bar line.                                                                                                                              |
| knob sensitivity     | How far faders move per encoder step (useful on slow-turning knobs).                                                                                                             |
| end warning          | How long before the end of a song it warns you (platter outline or waveform header flashes): off, 10, 20, 30 (default) or 60 seconds.                                            |
| clock follows master | Sets the norms clock tempo from the master deck, so clocked gear follows.                                                                                                        |
| panic: all notes off | Sends note-offs and all-notes-off on both decks' outputs.                                                                                                                        |
| screen view          | <b>platters</b> (default) or <b>waveform</b> .                                                                                                                                   |
| waveform zoom        | How much time the waveform shows across the screen: 2, 4 (default), 8, 16 or 32 seconds.                                                                                         |
| show frame time      | Shows how long each screen frame takes to draw, for checking the load on your norms.                                                                                             |
| clear song cache     | Deletes the song library's cached data. Cue points are kept.                                                                                                                     |

### Routing: one module or two

**One General MIDI module (the default).** Plug your GM module into the norms and set deck A's midi device to it; deck B plays to the same device automatically. General MIDI has only 16 channels, so deck B's channel shift (8 by default) moves its parts out of deck A's way: deck B's channels 1–7 play on 9 and 11–16, while deck A keeps 1–8. Both decks share drums on channel 10, because General MIDI has only one drum channel. Files that use more than 8 channels will still overlap; shared channels are handled for you (see How volume works).

**Two General MIDI modules.** Give each deck its own module, so nothing collides and both files keep all 16 channels.

- Plug both USB MIDI modules (or USB MIDI interfaces) into the norms.

- Go to SYSTEM › DEVICES › MIDI and check which port each one is on.
- In PARAMS › EDIT › MIDARTO, set **midi outputs** to two devices. Deck B's midi device setting appears.
- Set deck A's midi device to the first module and deck B's to the second.
- Set deck B's channel shift to 0.

*Upgrading from v2.3–v2.7: from 2.14 on, Midarto fixes deck B's old channel shift (0) for you on the first run and shows "B shift 8".*

## 3 Playing from the norns

The three knobs and three keys cover mixing and transport, so you can play a full set without a grid.

| Control | Normal            | Hold K1                          |
|---------|-------------------|----------------------------------|
| E1      | Crossfader        | Tempo of both decks              |
| E2      | Deck A volume     | Deck A tempo (0.1% steps)        |
| E3      | Deck B volume     | Deck B tempo                     |
| K2      | Deck A play/pause | Open the file browser for deck A |
| K3      | Deck B play/pause | Open the file browser for deck B |

*A quick tap on K1 still opens the norns system menu, as it does in every script, so K1 only works as a held shift key.*

## 4 The screen



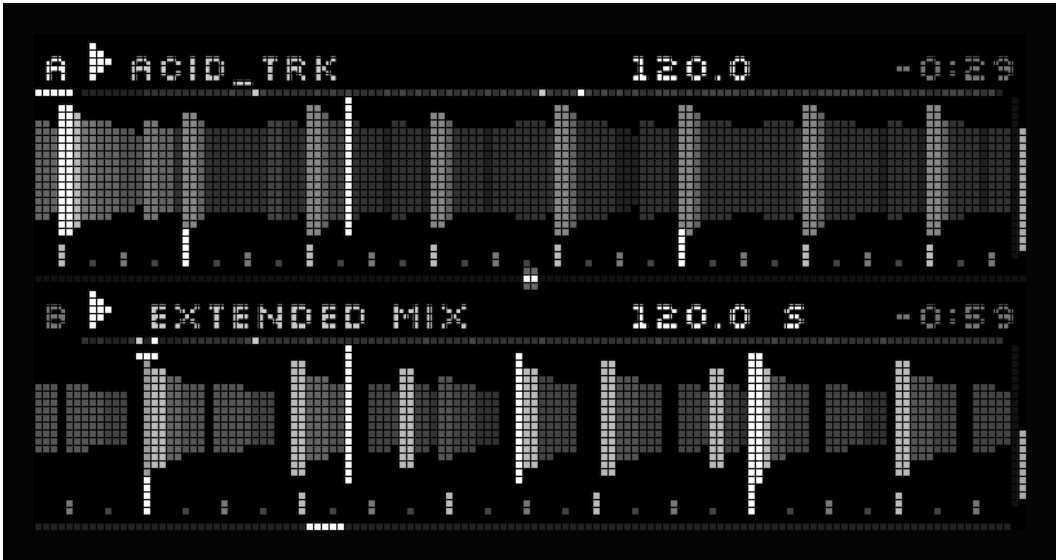
*The main screen: deck A on the left, the mixer in the middle, deck B on the right.*

| Element           | Meaning                                                                                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File names        | Top corners. Long names scroll: they pause at the start, scroll to the end, pause, then start again. Shows LOADING while a file is being read.                                                                       |
| Platters          | Turn once per bar at the deck's tempo. The marker is bright while playing and dims when paused; the tonearm lifts off when a deck stops. Deck B's label is lighter so you can tell them apart.                       |
| End warning       | When a playing song is near its end, its platter outline flashes: about once a second, then twice as fast in the last 10 seconds. It stays off while the deck is looping. Set the timing with end warning in PARAMS. |
| MASTER A / B      | Which deck is the tempo master.                                                                                                                                                                                      |
| Faders and meters | Deck A and B volume faders with output meters between them. An X under a fader means that deck is killed.                                                                                                            |
| Crossfader        | The horizontal fader below the mixer.                                                                                                                                                                                |
| Progress bars     | Position in the file. A tall tick marks hot cue 1, and a short line above shows the loop region.                                                                                                                     |
| BPM and tempo     | Current BPM, and the tempo offset in percent. The offset blinks while the deck is synced.                                                                                                                            |
| Time and state    | Time remaining, and a play or pause symbol.                                                                                                                                                                          |

| Element     | Meaning                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------|
| Status line | MIDARTO normally; SHIFT while K1 is held; SYNC when a deck is synced; short messages such as PANIC. |

Waveform view

Set **screen view** to waveform in PARAMS. Each deck gets half the screen, A on top and B below.



The waveform view: deck A (master) playing, deck B synced, both scrolling past the fixed playhead.

| Element        | Meaning                                                                                                                                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Header         | Deck letter (bright with a bar under it on the master deck), play/pause (blinking while armed), the scrolling song name, BPM (shows the tempo change for a second when you change it), a blinking S when synced, and time remaining.                                                                                                                |
| Overview strip | A thin line showing the whole song, the loop, hot cues and your position. A pending jump's target blinks.                                                                                                                                                                                                                                           |
| Waveform       | The song's MIDI activity scrolling past a fixed playhead. Height shows how busy the music you can hear is; kicks punch out as tall bright spikes and snares or claps as shorter light-grey ones. Songs without drums use bass-note starts. Upcoming music is slightly dimmer, the end-warning stretch darker, and muted channels disappear from it. |
| Markers        | Hot cues as lines, a dashed line where a pending jump will land, and a beat ruler: phrases every 8 bars, bars, beats, half and quarter beats.                                                                                                                                                                                                       |
| Right edge     | Each deck's output level.                                                                                                                                                                                                                                                                                                                           |
| Middle line    | Beat-phase meter: the marker sits on the centre mark when both decks' beats line up.                                                                                                                                                                                                                                                                |
| Bottom edge    | The crossfader.                                                                                                                                                                                                                                                                                                                                     |

The waveform is a picture of the MIDI notes, not the sound from your GM module, but it shows beats, drops and breakdowns clearly. Kick and snare detection uses the General MIDI drum map (kick on notes 35–36, snares and claps on 37–40).

### Song library

Midarto keeps a small library in `dust/data/midarto/library/`.

- **Cue points:** hot cues 1–4 and the last loop length for each song are remembered and come back when you load it again.
- **Cache:** the prepared song and its waveform, so loading a song the second time is much quicker. If you edit or replace a MIDI file, Midarto notices and rebuilds it, keeping your cue points if they still fit the song.
- Everything in the library is safe to delete. **clear song cache** in PARAMS removes the cached data but keeps your cue points.

## 5 The file browser

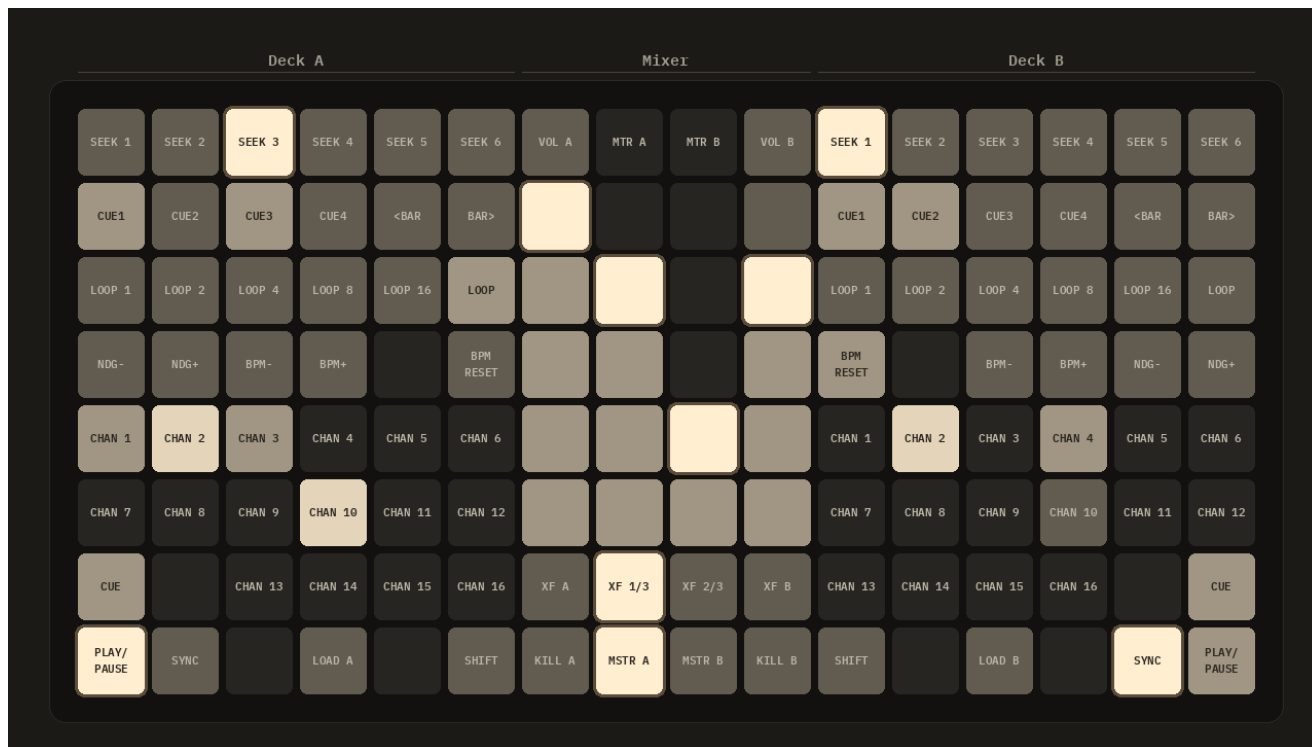
Hold K1 and press K2 (deck A) or K3 (deck B), or press LOAD on the grid. Press the same again to close the browser without loading anything, in case you opened it by accident. The other deck keeps playing while you browse.

| Control | In the browser                           |
|---------|------------------------------------------|
| E1      | Switch the target deck                   |
| E2      | Scroll the list                          |
| E3      | Scroll a page at a time                  |
| K3      | Open a folder, or load the selected file |
| K2      | Close the browser                        |

*Loading a file stops that deck first. Files are read in small slices so the other deck keeps playing smoothly; big files take a second or two, and the name shows `LOADING` until they're ready.*

## 6 The grid

Midarto uses a 16×8 grid. Deck A is on columns 1–6, the mixer on 7–10 and deck B on 11–16. Rows 1–3, 5 and 6 run in the same order on both decks. The tempo row and the bottom two rows are mirrored, so NDG, CUE, PLAY/PAUSE and SYNC sit on the outer edges under each hand; minus/plus pairs still read left to right.



The grid mid-set: deck A playing, deck B synced with its drums muted, crossfader towards A.

### Deck keys

| Row | Deck A (x1–x6)                              | Deck B (x11–x16)                            |
|-----|---------------------------------------------|---------------------------------------------|
| 1   | SEEK 1–6                                    | SEEK 1–6                                    |
| 2   | CUE1–4, <BAR, BAR>                          | CUE1–4, <BAR, BAR>                          |
| 3   | LOOP 1, 2, 4, 8, 16, LOOP                   | LOOP 1, 2, 4, 8, 16, LOOP                   |
| 4   | NDG-, NDG+, BPM-, BPM+, free, BPM RESET     | BPM RESET, free, BPM-, BPM+, NDG-, NDG+     |
| 5   | CHAN 1–6                                    | CHAN 1–6                                    |
| 6   | CHAN 7–12                                   | CHAN 7–12                                   |
| 7   | CUE, free, CHAN 13–16                       | CHAN 13–16, free, CUE                       |
| 8   | PLAY/PAUSE, SYNC, free, LOAD A, free, SHIFT | SHIFT, free, LOAD B, free, SYNC, PLAY/PAUSE |

### What the deck keys do

- **SEEK 1–6** jumps to that sixth of the file.
- **CUE1–4 (hot cues)**: press an empty one to store the current bar; press a stored one to jump there. SHIFT + hot cue erases it.
- **<BAR / BAR>** jump back or forward one bar.

- **LOOP 1–16** repeats that many bars from the current bar; press the lit one again to release. **LOOP** turns the last loop length back on or off.
- **NDG- / NDG+** nudge slower or faster by 4% while held, to line up beats by ear.
- **BPM- / BPM+** change tempo by 0.5%. With **SHIFT** held they transpose by a semitone instead (drums are never transposed). **BPM RESET** returns to the file tempo straight away and releases **SYNC**, or resets the transpose with **SHIFT**.
- **SYNC** (next to **PLAY/PAUSE**) matches the other deck's tempo, lines up the bar position and keeps following. Press again to release. The master deck never follows.
- **CUE** goes to hot cue 1 (or the start). While the deck is playing it jumps there and keeps playing, on the next bar when quantize jumps is on; while paused it moves there and stays paused. **SHIFT + CUE** stops the deck and returns to the start.
- **PLAY/PAUSE** starts or pauses the deck, and pulses on each beat. With start on bar on and the other deck playing, **PLAY** arms the deck: the key blinks and the deck starts exactly on the other deck's next bar line, synced with on + sync. Press **PLAY** again to cancel.
- **LOAD** opens the file browser for that deck; press it again to close it. **SHIFT** is shared by both decks; there's one next to the mixer on each side.
- **free** keys do nothing yet.

## Channel mute and solo

Each deck has a key for every MIDI channel, CHAN 1–16. These are the file's own channel numbers, so CHAN 10 is the drums even if you use channel shift.

- Press a **CHAN** key to mute or unmute that channel.
- **SHIFT + CHAN** solos that channel, so only soloed channels play. **SHIFT +** more channels adds them to the solo, for example drums plus bass.
- **SHIFT + a soloed channel** restores all channels, clearing every solo and mute.
- Every channel key flickers brighter each time its channel plays a note, so you can see which parts are active.

| Channel state                              | Light                                |
|--------------------------------------------|--------------------------------------|
| Not used in the file                       | Off                                  |
| Muted, or silent because others are soloed | Dim                                  |
| Playing                                    | Medium, flickering brighter on notes |
| Soloed                                     | Bright, still flickering on notes    |

## Mixer keys

| Row | x7            | x8           | x9           | x10           |
|-----|---------------|--------------|--------------|---------------|
| 1–6 | deck A volume | deck A meter | deck B meter | deck B volume |
| 7   | crossfader: A | 1/3          | 2/3          | B             |
| 8   | kill A        | master A     | master B     | kill B        |

- **Volume:** press a row to set the level (top row = 100%).
- **Meters** show each deck's output after volume, crossfader, kill and mutes. Hold a meter key to cut that deck until you let go.
- **Crossfader:** pressing a lit middle key moves the crossfader to the centre. Use E1 on the norms for smooth fades.

- **Kill** mutes a deck's whole output. **Master** sets the tempo reference for sync and for clock follows master.
- **Panic**: press master A and master B together.

## Light levels

| Brightness | Meaning       |
|------------|---------------|
| Off        | Nothing here  |
| Dim        | Available     |
| Medium     | Set or stored |
| Bright     | On or active  |

*PLAY pulses on each beat, and a key blinks while its jump waits for the next bar line.*

## 7 How volume works

For MIDI outputs, Midarto sets each channel's volume (MIDI CC7) from the file's own channel volume, multiplied by the deck fader, the crossfader and kill. Moving a fader therefore fades notes that are already sounding, and each file keeps its own mix balance. Updates are limited to about one every 30 ms per channel so the MIDI line doesn't flood.

For nb voices there's no channel volume, so Midarto scales note velocity instead. That only affects new notes, so long held notes won't fade with the fader.

## Instruments

When a file never chooses an instrument for a channel, Midarto sets piano (the General MIDI default), so the part doesn't pick up whatever instrument was left on that channel. Channel shift moves each part's instrument with it.

## Shared channels

When both decks use the same channel on the same device (always true for drums on channel 10 with one GM device), the playing deck owns it. Loading, cueing or stopping the other deck sends nothing to that channel: no volume, instrument, controller or all-notes-off messages, so the playing deck carries on untouched.

When both decks play, the louder one owns the channel and sets its volume and instrument, and the quieter deck's notes there play at its own level through velocity. Crossfading therefore hands the drums over from one deck to the other.

## 8 Tips for a first mix

- Load `acid_trk` into deck A and breaks into deck B, and start deck A.
- Press **PLAY** on deck B at any point: it skips any silence at the start, waits for deck A's next bar and comes in on the downbeat, locked to deck A's tempo.
- Turn E1 slowly to crossfade from A to B, watching the meters.
- Try muting deck B's drums (**CHAN 10**), or hold SHIFT and press deck A's bass channel to solo it, to layer the two tracks.
- Set a 4-bar **loop** on the outgoing deck to hold a groove while you bring the next one in.
- Store **hot cues** at the drops you like, so you can jump back to them on the bar.

## 9 Troubleshooting and limits

| Problem                 | Try this                                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| No sound                | Check the deck's midi device in PARAMS, the fader and crossfader, and that the deck isn't killed or cut.                         |
| Wrong instruments       | Some modules need a General MIDI reset: power-cycle it or enable its GM mode. Midarto re-sends program changes after every jump. |
| Decks fight over sounds | Both decks are on the same channels of one module. Give deck B a channel shift, or use two modules.                              |
| Stuck notes             | Run panic from PARAMS, or press master A and master B together on the grid.                                                      |
| A file won't load       | The status line shows the reason. Check the file is a standard MIDI file (format 0 or 1).                                        |
| Errors                  | Open maiden and look for REPL messages starting with midarto.                                                                    |

### Known limits in v2.14

- Reads MIDI file formats 0 and 1 and follows tempo changes inside the file. Files timed in SMPTE frames play at the right speed but show 60 BPM.
- Bars come from the file's first time signature (4/4 if there is none).
- Each deck's nb voice plays every channel of the file on that one voice.
- The grid layout is built for a 16×8 grid; other sizes aren't supported yet.

## Credits and contact

Midarto was created by **Glenn Jones, Cutie Suzuki & DJ FingaBlasta**.

|            |                                 |
|------------|---------------------------------|
| Email      | mailmrg@gmail.com               |
| Social     | @mrglennjones                   |
| GitHub     | github.com/mrglennjones         |
| Project    | github.com/mrglennjones/midarto |
| Discussion | 11111111.co/t/midarto/75532     |

Questions, bug reports and mix recordings are welcome. For bugs, include the REPL lines from maiden that start with midarto, and what you pressed.

### Thanks

- **monome** (brian crabtree and the norns contributors) for norns, maiden and the grid.
- **sixolet** for nb (nota bene), which Midarto can play through.
- **sonoCircuit** for midiplayer, which showed MIDI file playback on norns and inspired Midarto.
- **adamstaff** for turntable, whose norns DJ controls inspired Midarto's keys and knobs.
- **denki oto** for the norns Shield XL and the grid clone that Midarto was developed and tested on.

*This guide uses IBM Plex Mono, and its images use Silkscreen, both under the SIL Open Font License.*

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